These energy requirements of quasars and radio galaxies were so staggering that they forced astrophysicists, in 1963, to examine all conceivable sources of power in search of an explanation.

Chemical power (the burning of gasoline, oil, coal, or dynamite), which is the basis of human civilization, was clearly inadequate. The chemical efficiency for converting mass into energy is no larger than 1 part in a billion (1 part in 10^9). To energize a quasar's radio-emitting gas would therefore require $10^9 \times 10^7 = 10^{16}$ solar masses of chemical fuel—100,000 times more fuel than is contained in our entire Milky Way galaxy. This seemed totally unreasonable.

Nuclear power, the basis of the hydrogen bomb and of the Sun's heat and light, looked only marginal as a way to energize a quasar. Nuclear fuel's efficiency for mass-to-energy conversion is roughly 1 percent (1 part in 10^2), so a quasar would need $10^2 \times 10^7 = 10^9$ (1 billion) solar masses of nuclear fuel to energize its radio-emitting lobes. And this 1 billion solar masses would be adequate only if the nuclear fuel were burned completely and the resulting energy were converted completely into magnetic fields and kinetic energy of high-speed electrons. Complete burning and complete energy conversion seemed highly unlikely. Even with carefully contrived machines, humans rarely achieve better than a few percent conversion of fuel energy into useful energy, and nature without careful designs might well do worse. Thus, 10 billion or 100 billion solar masses of nuclear fuel seemed more reasonable. Now, this is less than the mass of a giant galaxy, but not a lot less, and how nature might achieve the conversion of the fuel's nuclear energy into magnetic and kinetic energy was very unclear. Thus, nuclear fuel *was* a possibility, but not a likely one.

The annihilation of *matter with antimatter*[3] could give 100 percent conversion of mass to energy, so 5 million solar masses of antimatter annihilating with 5 million solar masses of matter could satisfy a quasar's energy needs. However, there is no evidence that any antimatter exists in our Universe, except tiny bits created artifically by humans in particle accelerators and tiny bits created by nature in collisions between matter particles. Moreover, even if so much matter and antimatter were to annihilate in a quasar, their annihilation energy would go into very high